

MAT 310, Handout 23: Conditional Expectation (5.11)

Conditional Expectation

- Y_1, Y_2 Discrete:

$$E[g(Y_1)|Y_2 = y_2] =$$

- Y_1, Y_2 Continuous:

$$E[g(Y_1)|Y_2 = y_2] =$$

- If y_2 is a constant, then $E[g(Y_1)|Y_2 = y_2]$ is a number.
- If y_2 is a variable, then $E[g(Y_1)|Y_2 = y_2]$ is a function of y_2 .

Example 1. The following table gives the joint probabilities of Y_1 and Y_2 .

		Y ₂		
		0	1	2
Y ₁	0	1/9	2/9	1/9
	1	2/9	2/9	0
	2	1/9	0	0

$$E[Y_1^2|Y_2 = 0] =$$

Theorem (Law of Iterated Expectations). For any 2 RVs X and Y ,

$$E[E(Y|X)] =$$

Proof (continuous case):

Theorem (Law of Total Variance). For any 2 RVs X and Y ,

$$V(Y) = E[V(Y|X)] + V[E(Y|X)].$$

Example 2. The joint density of Y_1 and Y_2 is given by

$$f(y_1, y_2) = \begin{cases} 2, & 0 \leq y_1 \leq y_2 \leq 1 \\ 0, & o.w. \end{cases}.$$

Find the following:

(a) marginal density function of Y_2

(b) $E(Y_1 | Y_2 = y_2)$

(c) $E\left(Y_1 | Y_2 = \frac{1}{2}\right)$

(d) $E(Y_1)$ (using part (b))

Example 3. Suppose X is the number of buses that come by a bus stop in a given hour and that X follows a Poisson distribution with mean 10. The bus stop serves two different bus routes (A and B), with $P(\text{a random bus is Route A})=0.4$. Assume that the buses that come by the bus stop are independent of each other. Find the mean and standard deviation of Y , the number of Route A buses that come by the stop in one hour.